

Jay Liao

Jie Hao Liao | liaojh1998.github.io
liaojh@google.com | liaojhw@gmail.com | (737) 333-1729

EDUCATION

UNIVERSITY OF TEXAS

MS IN COMPUTER SCIENCE

August 2016 - May 2022 | Austin, TX
GPA: 3.9450 / 4.0

UNIVERSITY OF TEXAS

BS IN COMPUTER SCIENCE WITH HIGH HONORS

August 2020 - May 2020 | Austin, TX
GPA: 3.9525 / 4.0

LINKS

Github:// liaojh1998
LinkedIn:// jiehaoliao

SKILLS

Programming

Python • C/C++ • Java • SQL • Javascript (Node.js) • HTML/CSS

Technical

PyTorch • pandas • React.js • Node.js • MySQL • Docker • Git • Jupyter Notebook • tmux • vim • LaTeX

COURSEWORK

GRADUATE

CS 395T: Topics in NLP
LIN 393: Discourse Processing
CS 395T: Grounded NLP
CS 395T: Spoken Lang. Technologies
CS 391R: Robot Learning
EE 381V: Advanced Topics in CV
CS 395T: Deep Learning Seminar
CS 394N: Neural Networks
CS 388C: Combinatorics and Graph Theory

UNDERGRADUATE

CS 378: Natural Language Processing
CS 378: F1/10 Autonomous Driving
CS 342: Neural Networks
CS 364D: Advanced Data Mining
CS 373: Software Engineering
CS 378: Multicore Operating Systems
CS 439: Operating Systems
CS 378: Virtualization
CS 429: Computer Architecture
CS 331: Algorithms and Complexity
CS 378: Big Data Programming
CS 314: Data Structures

EXPERIENCE

GOOGLE | SOFTWARE ENGINEER

Aug 2022 - Present | Seattle, WA

- Created 30 data pipelines and 7 near real-time dashboards for monitoring if Google Cloud GPU and TPU machines can schedule VMs for all customers.
- Built automated releases and integration tests on a termination telemetry pipeline to improve pipeline reliability and data integrity.

UNIVERSITY OF TEXAS AT AUSTIN | GRADUATE TA

Aug 2020 - Dec 2020 | Austin, TX

- Led discussions, office hours, and graded for CS 331 Algorithms and Complexity with **Professor Fares Fraij**.

UNIVERSITY OF TEXAS AT AUSTIN | UNDERGRADUATE TA

Sep 2019 - May 2020 | Austin, TX

- Graded and proctored for CS 331 Algorithms and Complexity with **Professor Fares Fraij**.

META PLATFORMS | PERFORMANCE AND CAPACITY ENGINEER INTERN

May 2019 - Aug 2019 | Menlo Park, CA

- Implemented a pipeline that refreshed a general purpose profile for a profile-guided optimization, AutoFDO, using **Python**, which was found to reduce the CPU utilization of most binaries by $\sim 1\%$ and the capital expenditure by \$1-10 million.
- Implemented and shipped the UI and backend for the profiling of a memory allocator, jemalloc, using **React.js**, **Hack**, and **C++**, which assisted in identifying memory fragmentation with respect to time.

QUANTLAB FINANCIAL | SOFTWARE DEVELOPMENT INTERN

June 2018 - Aug 2018 | Houston, TX

- Implemented an automated anomaly detection pipeline for high-frequency trades using **Python**, which found failures on production machines that were left undiscovered for 6 months.

PROJECTS

REPRESENTING KNOWLEDGE BY GENERATING SENTENCES WITH RELATIONAL CONTEXTS

LIN 393 Discourse Processing, CS 395 Topics in NLP | [Paper Link](#)

Worked with **Ryo Kamoi**, **Prof Eunsol Choi**, and **Prof Jessy Li** to create a Transformer question answering model trained on common sense sentences with common sense information to answer common sense questions using **PyTorch**.

NEAR UNSUPERVISED TEXT-TO-SPEECH AND AUTOMATIC SPEECH RECOGNITION

CS 395T Speech Language Technologies | [Paper Link](#)

Worked with **Lucas Kabela**, and **Bill Yang**, and **Prof David Harwath** to create a semi-supervised Transformer model to do ASR and TTS conversion using **PyTorch**.

VOTEWISELY | FULL-STACK ENGINEER

CS 373 Software Engineering | [Project Link](#)

Engineered a web application in a team that provided citizens with unbiased information on recent controversial issues by using **React.js** for UI, **Express.js** and **MySQL** for the backend, and **Docker** on **GitLab** for continuous deployment.